# ADITHYAN R S

AI Enthusiast, Systems Thinking & Applied Machine Learning
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## SUMMARY

AI enthusiast with a strong systems-level understanding of modern machine learning architectures, including retrieval-augmented generation, LLM agents, and mechanistic interpretability. Combines deep conceptual grounding in how these systems work with AI-assisted engineering to build and iterate on complex projects efficiently, while developing hands-on programming fundamentals in parallel.

## SKILLS

**Core Programming:** Python, C | **Learning:** C++ (basic), C# (basic)

**AI / ML Concepts:** Retrieval-Augmented Generation, LLM agents, Prompt Engineering, mechanistic interpretability (sparse autoencoders, activation steering, model editing), classical ML (regression, gradient boosting), Deep Learning, Transformer Architecture, Computer Vision

**Familiar With:** LangGraph, FAISS, PyTorch, TransformerLens, SAELens, Flask, Streamlit, PyQt6, Groq API, Sarvam AI, Telegram Bot API

**CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems

## PROJECTS

**Transient Activation Steering for Language Models** *Python, PyTorch, TransformerLens, SAELens, Streamlit, CUDA* github.com/adi5022/Mechanistic-Intervention...

- Built an interactive research platform for inference-time intervention in GPT-2 Small using a pretrained sparse autoencoder (SAE), editing the residual stream through SAE features for one forward pass, with no retraining or weight updates.
- Implemented gradient-based optimization of one multiplier per SAE feature; on 300 held-out CounterFact prompts it brought the true answer to rank 1 on 271 (90.3%), versus 161 (53.7%) for a fixed-strength sweep given the same range.
- Cut model time to 7.7 s per prompt versus 38.5 s for the wider sweep (about 5x faster) and measured side effects with KL divergence; batched implementation reduced model calls from 186 to 6 per prompt and layer.
- Ran controls on its own results: an additive edit forced random words to rank 1 on 20 of 20 prompts, so rank 1 alone is a weak test; reproduced ROME as an outside reference. Cross-model transfer to GPT-2 Medium moved the median target rank from 142 to 14 but reached rank 1 on only 8% of records.

**Sahayi: Malayalam-First Care Assistant** *Python, Groq, Sarvam AI, SQLite, Telegram, React* github.com/adi5022/sahayiAI

- Built with a three-person team at the BIA GenAI Hackathon 2026: a Telegram assistant for seniors and people with disabilities that takes voice, photo or text, rates urgency from 1 to 3, and alerts a linked family member, without ever contacting emergency services itself.
- Routed Malayalam and code-mixed speech to Sarvam Saaras and English to Groq Whisper with no silent fallback between them; added a two-model photo check reconciled by a judge model, and a care agent returning Pydantic-validated JSON.
- Designed separate working memory and long-term history in SQLite, plus scam-help flows, family Q&A and relay commands, and a browser simulator for testing.

**MBAssist AI: Multilingual RAG Admissions Assistant** *Python, Flask, LangGraph, FAISS, Groq (Llama-3.3-70B), Sarvam AI* github.com/adi5022/MBAssist-AI

- Built a voice-enabled, multi-agent RAG chatbot for college admissions, using LangGraph-orchestrated retrieval over scraped institutional documents and a FAISS vector index.
- Designed a hybrid memory architecture, sliding-window recent context plus LLM-summarized long-term archive, to bound token usage; built English and Malayalam speech-to-text via Sarvam AI and Groq Whisper, with a per-user API-key model that keeps hosting costs at zero.
- Set up automated weekly re-indexing of the source website through a background scheduler with thread-safe rebuild locks.

**Tarang AI: Voice and Text Sentiment Bot** *Python, Groq Whisper, Sarvam AI, wav2vec2, Streamlit, Telegram* github.com/adi5022/tarangAI

- Built a Telegram bot that reads sentiment in English and Malayalam-English code-mixed voice notes and text, routing by language and running sentiment on the original text with no translation step.
- Added a fusion layer for English voice that combines a wav2vec2 arousal and valence model with text sentiment, adjusting intensity by one step and flagging sarcasm-like mismatches; voice cut-offs calibrated from recorded clips (one speaker, no formal evaluation).

**GraphRAG Research Assistant: Hybrid Vector + Graph RAG** *Python, Flask, FAISS, NetworkX, Groq/Gemini* github.com/Tharun-1-6/graph-rag-research-assistant

- Co-built a hybrid RAG system for scientific literature, combining a FAISS vector index with a NetworkX knowledge graph; designed the query routing that selects vector, graph, or combined retrieval, and built the interactive graph and retrieval-trace layer.

**FlowState AI: Cardiovascular Digital Twin** *Python, PyQt6, scikit-learn, Plotly, Gemini/Groq* github.com/adi5022/flow-state-AI

- Designed a patient-specific digital twin projecting blood pressure, kidney function (eGFR) and survival over 30 years from 10 clinical inputs, with a two-layer eGFR model (clinical equations plus gradient boosting residual) at cross-validated $R^2$ of 0.957; part of a 4-person team.

## EDUCATION

**B.Tech, Computer Science and Engineering (AI)** 2023 to 2027
Mar Baselios College of Engineering & Technology, Thiruvananthapuram

## CERTIFICATIONS

**AI Agent Architect:** IBM SkillsBuild, Aug 2025 | **CS50 Intro to Python:** Harvard (edX), May 2024 | **Flutter App Development:** Infosys Springboard, Oct 2024